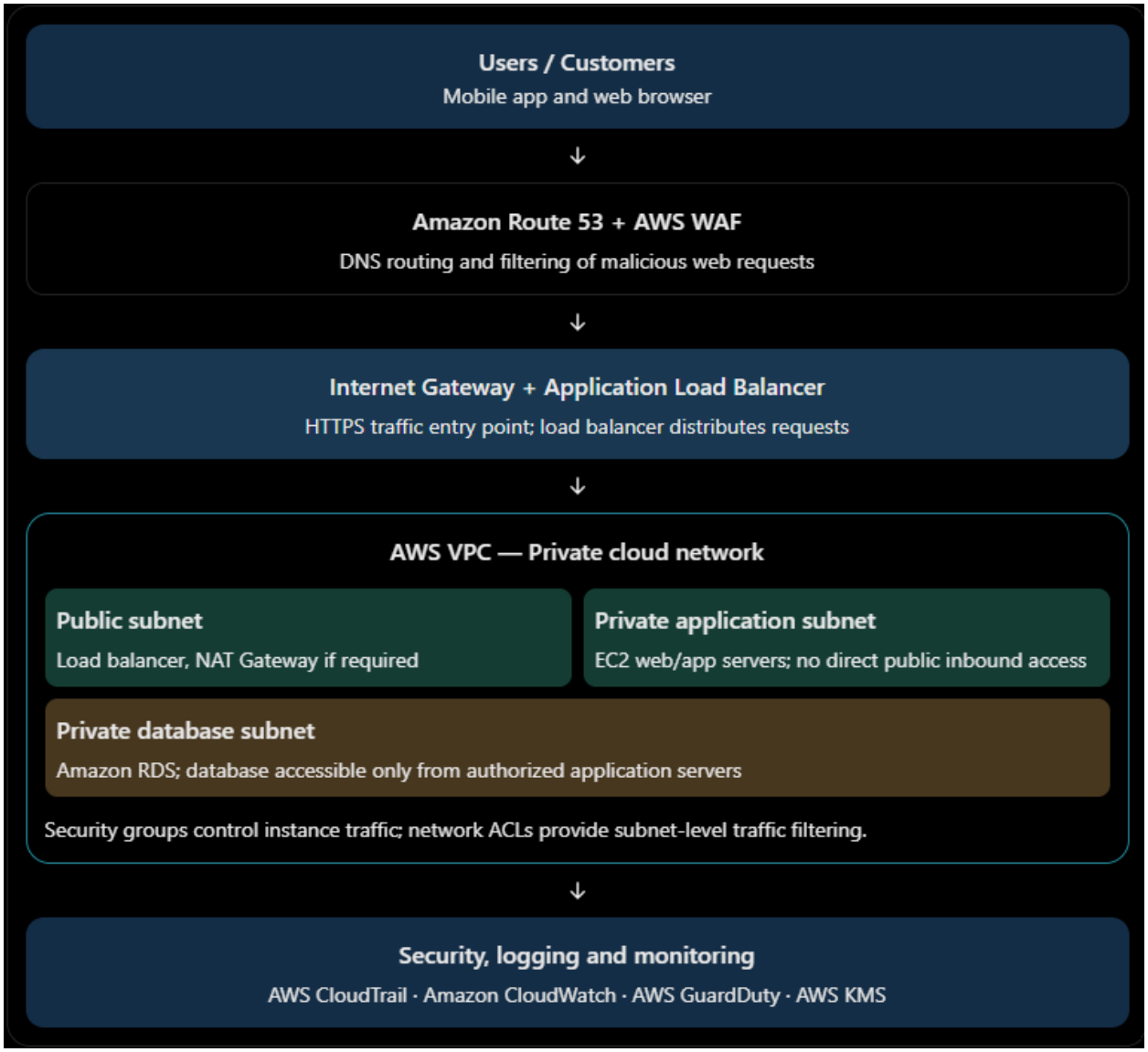
Task 1: Secure cloud architecture for an online food delivery app



Components and their security purposes

Component	Security purpose
Amazon Route 53	Directs users to the correct application endpoint using DNS.
AWS WAF	Filters malicious web traffic, including common SQL injection and cross-site scripting attempts.

Application Load Balancer	Distributes application traffic and supports HTTPS termination.
Amazon VPC	Isolates the app's cloud network and controls network connectivity.
Public subnet	Hosts components that need a controlled internet-facing entry point.
Private application subnet	Keeps application servers inaccessible to direct public inbound connections.
Private database subnet	Protects customer details, addresses, orders and payment-related records from direct internet access.
Security groups and network ACLs	Restrict traffic according to source, destination, port and protocol.
AWS IAM	Grants users and services only the permissions they need.
AWS KMS	Manages encryption keys used to protect stored data.
CloudWatch and CloudTrail	Monitor system activity, metrics, logs and API actions.
GuardDuty	Detects suspicious activity and potential threats in AWS accounts and workloads.

Task 2: Five key security controls for a music streaming app like Spotify

Security control	How it protects user data

1. IAM and least privilege	Limits access to customer profiles, playlists and storage according to each user's or service's role.
2. Multi-factor authentication (MFA)	Adds a second verification factor to reduce the risk of account compromise from stolen passwords.
3. Encryption at rest and in transit	Uses encryption such as AES-256 for stored data and TLS for data moving between clients and servers.
4. Web application firewall (WAF)	Helps block malicious requests, automated attacks and common web application exploits.
5. Logging, monitoring and alerting	Detects suspicious login attempts, unusual API activity and unexpected changes to cloud resources.

Task 3: Comparison of cloud monitoring tools

Feature	AWS CloudWatch	Azure Monitor	Google Cloud Operations Suite
Cloud provider	Amazon Web Services	Microsoft Azure	Google Cloud
Metrics	CPU, network, resource and application metrics	Resource and application metrics	Infrastructure and application metrics
Alerting	Metric alarms and log-based alarms	Metric, log and activity alerts	Alerting policies and notification channels

Dashboards	Custom dashboards for metrics and logs	Azure dashboards, Workbooks and Grafana integrations	Built-in and custom dashboards
Log analysis	CloudWatch Logs Insights	Log Analytics with KQL	Cloud Logging and Logs Explorer
Main benefit	Integrated monitoring for AWS resources	Monitoring for Azure and hybrid environments	Monitoring across Google Cloud resources and supported integrations

Task 4: Incident response workflow for a payment app like Paytm

1 Detection

Monitoring tools detect suspicious logins, unusual transactions or unexpected data access.

2 Triage and investigation

The security team checks CloudTrail events, application logs, affected accounts and the severity of the incident.

3 Containment

Restrict compromised IAM credentials, isolate affected instances and block malicious traffic without unnecessarily disrupting unaffected services.

4 Eradication

Remove malicious code, close exploited vulnerabilities and rotate exposed credentials or keys.

5 Recovery

Restore trusted services, verify data integrity and monitor the application for recurring suspicious activity.

6 Review and prevention

Document the incident, notify the appropriate stakeholders, preserve evidence and improve security controls.

Alerts that can trigger the workflow

- GuardDuty finding: Suspicious API activity or behavior associated with a potential compromise.
- CloudTrail alert: Unexpected IAM policy changes, unusual API calls or attempts to disable logging.
- CloudWatch alarm: Sudden CPU or network spikes, repeated application failures, or unusual error rates.
- AWS WAF alert or logs: A high volume of malicious requests or repeated web attack attempts.
- Application security alert: Unusual payment attempts, repeated authentication failures or suspicious access to customer records.

Task 5: Security risks and monitoring strategies for Google Drive

Google Drive may face risks such as unauthorized access if an account password is stolen and accidental exposure when files are shared with the wrong people. One possible monitoring strategy is detecting suspicious sign-in activity, including unusual login patterns and potentially compromised accounts. Another is auditing file-sharing changes and access events to identify unexpected public links or unusual file downloads.

Multi-factor authentication and restricted sharing permissions can reduce these risks. The exact internal monitoring rules and detection thresholds used by Google are not fully public.